

MYSQL VIEWS

Practical Examples and Student Exercises using MySQL Workbench

2520030395

MONISH PUTTI

1.What is a View?

A **View** is a **virtual table** created from one or more existing tables.

- It **does not store data** itself.
- It stores only the **SQL query**.
- Whenever you query the view, MySQL retrieves the latest data from the underlying tables.

2. Setup: Employees and Departments Tables

Run this setup first. If these tables already exist with data, you may skip the setup.

```
CREATE DATABASE IF NOT EXISTS employeeedb1;  
USE employeeedb1;
```

```
CREATE TABLE IF NOT EXISTS departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE IF NOT EXISTS employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    dept_id INT,  
    salary DECIMAL(10,2)  
);
```

```
INSERT IGNORE INTO departments VALUES  
(1,'CSE'),  
(2,'ECE'),  
(3,'EEE');
```

```
INSERT IGNORE INTO employees VALUES  
(101,'Rahul',1,50000),  
(102,'Priya',2,65000),  
(103,'Anil',1,55000),  
(104,'Sneha',3,70000),  
(105,'Kiran',2,48000);
```

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 SHOW TABLES;
2
3 SELECT * FROM departments;
4
5 SELECT * FROM employees;

```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	51000.00
	102	Priya	3	67000.00
	103	Anil	1	57000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
*	NULL	NULL	NULL	NULL

Result 1 departments 2 employees 3 x Apply Revert Context Help Snip

Output

Action Output

#	Time	Action	Message
4	13:28:19	CREATE TABLE IF NOT EXISTS employees (...	0 row(s) affected, 1 warning(s): 1050 Tab
5	13:28:19	INSERT IGNORE INTO departments VALUES (1,...	0 row(s) affected, 3 warning(s): 1062 Dup
6	13:28:19	INSERT IGNORE INTO employees VALUES (101...	0 row(s) affected, 5 warning(s): 1062 Dup
7	13:28:46	SHOW TABLES	2 row(s) returned
8	13:28:46	SELECT * FROM departments LIMIT 0, 1000	3 row(s) returned
9	13:28:46	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned

3. Verify the Tables

SHOW TABLES;

SELECT * FROM departments;

SELECT * FROM employees;

View 1 – Display All Employees

Question: Create a View named EmployeeView to display all columns from employees.

DROP VIEW IF EXISTS EmployeeView;

CREATE VIEW EmployeeView AS
SELECT *
FROM employees;

SELECT * FROM EmployeeView;

The screenshot displays the SQL Developer interface. The top pane, titled 'Query 1', contains the following SQL code:

```
1 DROP VIEW IF EXISTS EmployeeView;
2
3 CREATE VIEW EmployeeView AS
4 SELECT *
5 FROM employees;
6
7 SELECT * FROM EmployeeView;
```

The middle pane shows the 'Result Grid' with the following data:

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	51000.00
	102	Priya	3	67000.00
	103	Anil	1	57000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00

The bottom pane, titled 'Output', shows the 'Action Output' log:

#	Time	Action	Message
7	13:28:46	SHOW TABLES	2 row(s) returned
8	13:28:46	SELECT * FROM departments LIMIT 0, 1000	3 row(s) returned
9	13:28:46	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned
10	13:31:11	DROP VIEW IF EXISTS EmployeeView	0 row(s) affected, 1 warning(s): 1051 Un
11	13:31:11	CREATE VIEW EmployeeView AS SELECT * FR...	0 row(s) affected
12	13:31:11	SELECT * FROM EmployeeView LIMIT 0, 1000	5 row(s) returned

View 2 – Selected Columns

Question: Create a View showing employee ID, employee name and salary only.

```
DROP VIEW IF EXISTS EmployeeBasicView;
```

```
CREATE VIEW EmployeeBasicView AS  
SELECT emp_id, emp_name, salary  
FROM employees;
```

```
SELECT * FROM EmployeeBasicView;
```

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS EmployeeBasicView;
2
3 CREATE VIEW EmployeeBasicView AS
4 SELECT emp_id, emp_name, salary
5 FROM employees;
6
7 SELECT * FROM EmployeeBasicView;

```

Result Grid

	emp_id	emp_name	salary
▶	101	Rahul	51000.00
	102	Priya	67000.00
	103	Anil	57000.00
	104	Sneha	70000.00
	105	Kiran	48000.00

EmployeeBasicView 5 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
10	13:31:11	DROP VIEW IF EXISTS EmployeeView	0 row(s) affected, 1 warning(s): 1051 Unl
11	13:31:11	CREATE VIEW EmployeeView AS SELECT * FR...	0 row(s) affected
12	13:31:11	SELECT * FROM EmployeeView LIMIT 0, 1000	5 row(s) returned
13	13:31:51	DROP VIEW IF EXISTS EmployeeBasicView	0 row(s) affected, 1 warning(s): 1051 Unl
14	13:31:51	CREATE VIEW EmployeeBasicView AS SELECT ...	0 row(s) affected
15	13:31:51	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned

View 3 – High Salary Employees

Question: Create a View that displays employees earning more than 55000.

DROP VIEW IF EXISTS HighSalaryEmployees;

```

CREATE VIEW HighSalaryEmployees AS
SELECT emp_id, emp_name, salary
FROM employees

```

WHERE salary > 55000;

SELECT * FROM HighSalaryEmployees;

The screenshot displays a SQL IDE interface. The top pane shows a query script for 'Query 1' with the following SQL statements:

```
1 DROP VIEW IF EXISTS HighSalaryEmployees;  
2  
3 CREATE VIEW HighSalaryEmployees AS  
4 SELECT emp_id, emp_name, salary  
5 FROM employees  
6 WHERE salary > 55000;  
7  
8 SELECT * FROM HighSalaryEmployees;
```

The bottom pane shows the 'Result Grid' with the following data:

emp_id	emp_name	salary
102	Priya	67000.00
103	Anil	57000.00
104	Sneha	70000.00

The bottom pane also shows the 'Output' window with the following log entries:

#	Time	Action	Message
13	13:31:51	DROP VIEW IF EXISTS EmployeeBasicView	0 row(s) affected, 1 warning(s): 1051 Un
14	13:31:51	CREATE VIEW EmployeeBasicView AS SELECT ...	0 row(s) affected
15	13:31:51	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned
16	13:32:22	DROP VIEW IF EXISTS HighSalaryEmployees	0 row(s) affected, 1 warning(s): 1051 Un
17	13:32:22	CREATE VIEW HighSalaryEmployees AS SELEC...	0 row(s) affected
18	13:32:22	SELECT * FROM HighSalaryEmployees LIMIT 0, ...	3 row(s) returned

View 4 – Employee and Department using JOIN

Question: Create a View that displays employee name, department name and salary.

DROP VIEW IF EXISTS EmployeeDepartmentView;

CREATE VIEW EmployeeDepartmentView AS
SELECT

 e.emp_id,
 e.emp_name,
 d.dept_name,
 e.salary

FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id;

SELECT * FROM EmployeeDepartmentView;

Query 1 x

SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS EmployeeDepartmentView;
2
3 CREATE VIEW EmployeeDepartmentView AS
4 SELECT
5     e.emp_id,
6     e.emp_name,
7     d.dept_name,
8     e.salary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id;
12

```

Result Grid

	emp_id	emp_name	dept_name	salary
▶	101	Rahul	CSE	51000.00
	102	Priya	EEE	67000.00
	103	Anil	CSE	57000.00
	104	Sneha	EEE	70000.00
	105	Kiran	ECE	48000.00

EmployeeDepartmentView 7 x

Read Only

Context Help

Output

Action Output

	#	Time	Action	Message
⚠	16	13:32:22	DROP VIEW IF EXISTS HighSalaryEmployees	0 row(s) affected, 1 warning(s): 1051 Unl
✓	17	13:32:22	CREATE VIEW HighSalaryEmployees AS SELEC...	0 row(s) affected
✓	18	13:32:22	SELECT * FROM HighSalaryEmployees LIMIT 0, ...	3 row(s) returned
⚠	19	13:32:59	DROP VIEW IF EXISTS EmployeeDepartmentView	0 row(s) affected, 1 warning(s): 1051 Unl
✓	20	13:32:59	CREATE VIEW EmployeeDepartmentView AS SE...	0 row(s) affected
✓	21	13:32:59	SELECT * FROM EmployeeDepartmentView LIM...	5 row(s) returned

View 5 – Employees from a Particular Department

Question: Create a View that displays only CSE employees.

DROP VIEW IF EXISTS CSEEmployees;

CREATE VIEW CSEEmployees AS
SELECT

```

e.emp_id,
e.emp_name,
d.dept_name,
e.salary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
WHERE d.dept_name = 'CSE';

SELECT * FROM CSEEmployees;

```

The screenshot displays the SQL Developer interface. The top pane shows the SQL script for creating and querying the CSEEmployees view. The middle pane shows the result grid with two rows of employee data. The bottom pane shows the output window with a list of actions and their messages.

SQL Script:

```

1 DROP VIEW IF EXISTS CSEEmployees;
2
3 CREATE VIEW CSEEmployees AS
4 SELECT
5     e.emp_id,
6     e.emp_name,
7     d.dept_name,
8     e.salary
9 FROM employees e
10 JOIN departments d
11 ON e.dept_id = d.dept_id
12 WHERE d.dept_name = 'CSE';

```

Result Grid:

emp_id	emp_name	dept_name	salary
101	Rahul	CSE	51000.00
103	Anil	CSE	57000.00

Output Window:

#	Time	Action	Message
19	13:32:59	DROP VIEW IF EXISTS EmployeeDepartmentView	0 row(s) affected, 1 warning(s): 1051 Unl
20	13:32:59	CREATE VIEW EmployeeDepartmentView AS SE...	0 row(s) affected
21	13:32:59	SELECT * FROM EmployeeDepartmentView LIM...	5 row(s) returned
22	13:33:32	DROP VIEW IF EXISTS CSEEmployees	0 row(s) affected, 1 warning(s): 1051 Unl
23	13:33:32	CREATE VIEW CSEEmployees AS SELECT e....	0 row(s) affected
24	13:33:32	SELECT * FROM CSEEmployees LIMIT 0, 1000	2 row(s) returned

View 6 – Department-wise Average Salary

Question: Create a View using GROUP BY and AVG() to display department-wise average salary.

```
DROP VIEW IF EXISTS DepartmentSalaryView;
```

```

CREATE VIEW DepartmentSalaryView AS
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS EmployeeCount,
    AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name;

SELECT * FROM DepartmentSalaryView;

```

The screenshot shows the SQL Developer interface. The top pane displays the SQL script for creating and querying the view. The bottom pane shows the results of the queries.

SQL Script:

```

1 DROP VIEW IF EXISTS DepartmentSalaryView;
2
3 CREATE VIEW DepartmentSalaryView AS
4 SELECT
5     d.dept_name,
6     COUNT(e.emp_id) AS EmployeeCount,
7     AVG(e.salary) AS AverageSalary
8 FROM employees e
9 JOIN departments d
10 ON e.dept_id = d.dept_id
11 GROUP BY d.dept_id, d.dept_name;
12

```

Result Grid:

dept_name	EmployeeCount	AverageSalary
CSE	2	54000.000000
EEE	2	68500.000000
ECE	1	48000.000000

Output Log:

#	Time	Action	Message
22	13:33:32	DROP VIEW IF EXISTS CSEEmployees	0 row(s) affected, 1 warning(s): 1051 Un
23	13:33:32	CREATE VIEW CSEEmployees AS SELECT e....	0 row(s) affected
24	13:33:32	SELECT * FROM CSEEmployees LIMIT 0, 1000	2 row(s) returned
25	13:34:35	DROP VIEW IF EXISTS DepartmentSalaryView	0 row(s) affected, 1 warning(s): 1051 Un
26	13:34:35	CREATE VIEW DepartmentSalaryView AS SELE...	0 row(s) affected
27	13:34:35	SELECT * FROM DepartmentSalaryView LIMIT 0,...	3 row(s) returned

View 7 – Departments with Average Salary Above 55000

Question: Create a View that displays departments whose average salary is greater than 55000.

```

DROP VIEW IF EXISTS HighAverageDepartments;

```

```

CREATE VIEW HighAverageDepartments AS
SELECT
    d.dept_name,

```

```

    AVG(e.salary) AS AverageSalary
FROM employees e
JOIN departments d
ON e.dept_id = d.dept_id
GROUP BY d.dept_id, d.dept_name
HAVING AVG(e.salary) > 55000;

SELECT * FROM HighAverageDepartments;

```

The screenshot shows the SQL Developer interface. The top pane displays the SQL script for creating and querying the `HighAverageDepartments` view. The script is as follows:

```

1 DROP VIEW IF EXISTS HighAverageDepartments;
2
3 CREATE VIEW HighAverageDepartments AS
4 SELECT
5     d.dept_name,
6     AVG(e.salary) AS AverageSalary
7 FROM employees e
8 JOIN departments d
9     ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name
11 HAVING AVG(e.salary) > 55000;
12

```

The bottom pane shows the results of the query. The `Result Grid` displays the following data:

dept_name	AverageSalary
EEE	68500.000000

The bottom pane also shows the `Output` window with the following messages:

#	Time	Action	Message
25	13:34:35	DROP VIEW IF EXISTS DepartmentSalaryView	0 row(s) affected, 1 warning(s): 1051 Un
26	13:34:35	CREATE VIEW DepartmentSalaryView AS SELE...	0 row(s) affected
27	13:34:35	SELECT * FROM DepartmentSalaryView LIMIT 0,...	3 row(s) returned
28	13:35:08	DROP VIEW IF EXISTS HighAverageDepartments	0 row(s) affected, 1 warning(s): 1051 Un
29	13:35:08	CREATE VIEW HighAverageDepartments AS SE...	0 row(s) affected
30	13:35:08	SELECT * FROM HighAverageDepartments LIM...	1 row(s) returned

View 8 – Salary Range View

Question: Create a View that displays employees with salary between 50000 and 70000.

```

DROP VIEW IF EXISTS SalaryRangeView;

CREATE VIEW SalaryRangeView AS
SELECT emp_id, emp_name, salary
FROM employees
WHERE salary BETWEEN 50000 AND 70000;

SELECT * FROM SalaryRangeView;

```

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS SalaryRangeView;
2 • CREATE VIEW SalaryRangeView AS
3 • SELECT emp_id, emp_name, salary
4 • FROM employees
5 • WHERE salary BETWEEN 50000 AND 70000;
6 •
7 •
8 • SELECT * FROM SalaryRangeView;

```

Set limit for number of rows returned by queries.
Workbench will automatically add the LIMIT clause with the configured number of rows to SELECT queries.

Result Grid

emp_id	emp_name	salary
101	Rahul	51000.00
102	Priya	67000.00
103	Anil	57000.00
104	Sneha	70000.00

SalaryRangeView 11 x Read Only Context Help Snip

Output

#	Time	Action	Message
28	13:35:08	DROP VIEW IF EXISTS HighAverageDepartments	0 row(s) affected, 1 warning(s): 1051 Un
29	13:35:08	CREATE VIEW HighAverageDepartments AS SE...	0 row(s) affected
30	13:35:08	SELECT * FROM HighAverageDepartments LIM...	1 row(s) returned
31	13:35:37	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected, 1 warning(s): 1051 Un
32	13:35:37	CREATE VIEW SalaryRangeView AS SELECT e...	0 row(s) affected
33	13:35:37	SELECT * FROM SalaryRangeView LIMIT 0, 1000	4 row(s) returned

View 9 – View with Calculated Column

Question: Create a View that displays employee name, monthly salary and annual salary.

DROP VIEW IF EXISTS AnnualSalaryView;

```

CREATE VIEW AnnualSalaryView AS
SELECT
    emp_id,
    emp_name,
    salary AS MonthlySalary,
    salary * 12 AS AnnualSalary
FROM employees;

```

SELECT * FROM AnnualSalaryView;

Query 1 x

SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS AnnualSalaryView;
2
3 CREATE VIEW AnnualSalaryView AS
4 SELECT
5     emp_id,
6     emp_name,
7     salary AS MonthlySalary,
8     salary * 12 AS AnnualSalary
9 FROM employees;
10
11 • SELECT * FROM AnnualSalaryView;

```

Result Grid

	emp_id	emp_name	MonthlySalary	AnnualSalary
▶	101	Rahul	51000.00	612000.00
	102	Priya	67000.00	804000.00
	103	Anil	57000.00	684000.00
	104	Sneha	70000.00	840000.00
	105	Kiran	48000.00	576000.00

AnnualSalaryView 12 x

Read Only

Context Help

Output

Action Output

#	Time	Action	Message
31	13:35:37	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected, 1 warning(s): 1051 Unl
32	13:35:37	CREATE VIEW SalaryRangeView AS SELECT e...	0 row(s) affected
33	13:35:37	SELECT * FROM SalaryRangeView LIMIT 0, 1000	4 row(s) returned
34	13:36:04	DROP VIEW IF EXISTS AnnualSalaryView	0 row(s) affected, 1 warning(s): 1051 Unl
35	13:36:04	CREATE VIEW AnnualSalaryView AS SELECT ...	0 row(s) affected
36	13:36:04	SELECT * FROM AnnualSalaryView LIMIT 0, 1000	5 row(s) returned

View 10 – Update Data Through a Simple View

Question: Create a simple single-table View and update an employee's salary through the View.

DROP VIEW IF EXISTS EmployeeSalaryView;

```

CREATE VIEW EmployeeSalaryView AS
SELECT emp_id, emp_name, salary
FROM employees;

```

```

UPDATE EmployeeSalaryView
SET salary = 60000
WHERE emp_id = 101;

```

```

SELECT * FROM EmployeeSalaryView;
SELECT * FROM employees WHERE emp_id = 101;

```

Query 1 x SQLAdditions

Limit to 1000 rows

```

3 CREATE VIEW EmployeeSalaryView AS
4 SELECT emp_id, emp_name, salary
5 FROM employees;
6
7 • UPDATE EmployeeSalaryView
8 SET salary = 60000
9 WHERE emp_id = 101;
10
11 SELECT * FROM EmployeeSalaryView;
12
13 • SELECT * FROM employees
14 WHERE emp_id = 101;

```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
*	NULL	NULL	NULL	NULL

Form Editor

EmployeeSalaryView 13 employees 14 x Apply Revert Context Help Snip

Output

Action Output

#	Time	Action	Message
✓ 36	13:36:04	SELECT * FROM AnnualSalaryView LIMIT 0, 1000	5 row(s) returned
⚠ 37	13:36:40	DROP VIEW IF EXISTS EmployeeSalaryView	0 row(s) affected, 1 warning(s): 1051 Un
✓ 38	13:36:40	CREATE VIEW EmployeeSalaryView AS SELECT...	0 row(s) affected
✓ 39	13:36:40	UPDATE EmployeeSalaryView SET salary = 6000...	1 row(s) affected Rows matched: 1 Cha
✓ 40	13:36:40	SELECT * FROM EmployeeSalaryView LIMIT 0, 1...	5 row(s) returned
✓ 41	13:36:40	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned

View 11 – CREATE OR REPLACE VIEW

Question: Modify an existing View so that it displays employee name and salary only.

```

CREATE OR REPLACE VIEW EmployeeBasicView AS
SELECT emp_name, salary
FROM employees;

```

```

SELECT * FROM EmployeeBasicView;

```

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • CREATE OR REPLACE VIEW EmployeeBasicView AS
2   SELECT emp_name, salary
3   FROM employees;
4
5 • SELECT * FROM EmployeeBasicView;

```

Automatically disabled. manually generated caret position auto

Result Grid | Filter Rows: | Export: | Write

	emp_name	salary
▶	Rahul	60000.00
	Priya	67000.00
	Anil	57000.00
	Sneha	70000.00
	Kiran	48000.00

Result Grid
Form Editor

EmployeeBasicView 15 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
✓ 38	13:36:40	CREATE VIEW EmployeeSalaryView AS SELECT...	0 row(s) affected
✓ 39	13:36:40	UPDATE EmployeeSalaryView SET salary = 6000...	1 row(s) affected Rows matched: 1 Cha
✓ 40	13:36:40	SELECT * FROM EmployeeSalaryView LIMIT 0, 1...	5 row(s) returned
✓ 41	13:36:40	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
✓ 42	13:37:10	CREATE OR REPLACE VIEW EmployeeBasicVie...	0 row(s) affected
✓ 43	13:37:10	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned

View 12 – View Definition

Question: Display the SQL statement used to create a View.

SHOW CREATE VIEW EmployeeDepartmentView;

Query 1 x

SQLAdditions

Limit to 1000 rows

1 • SHOW CREATE VIEW EmployeeDepartmentView;

Result Grid | Filter Rows: | Export: | Wrap Cells

View	Create View
employeedepartmentview	CREATE ALGORITHM=UNDEFINED DEFINER=

Result Grid

Form Editor

Result 16 x

Read Only

Context Help

Output

Action Output

#	Time	Action	Message
39	13:36:40	UPDATE EmployeeSalaryView SET salary = 6000...	1 row(s) affected Rows matched: 1 Cha
40	13:36:40	SELECT * FROM EmployeeSalaryView LIMIT 0, 1...	5 row(s) returned
41	13:36:40	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
42	13:37:10	CREATE OR REPLACE VIEW EmployeeBasicVie...	0 row(s) affected
43	13:37:10	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned
44	13:37:41	SHOW CREATE VIEW EmployeeDepartmentView	1 row(s) returned

View 13 – List All Views

Question: Display all Views available in the current database.

```
SHOW FULL TABLES
WHERE Table_type = 'VIEW';
```

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 SHOW FULL TABLES
2 WHERE Table_type = 'VIEW';

```

Result Grid | Filter Rows: | Export: | Wrap Cell

Tables_in_employeedb1	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
highsalaryemployees	VTFW

Result 17 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
✓ 40	13:36:40	SELECT * FROM EmployeeSalaryView LIMIT 0, 1...	5 row(s) returned
✓ 41	13:36:40	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
✓ 42	13:37:10	CREATE OR REPLACE VIEW EmployeeBasicVie...	0 row(s) affected
✓ 43	13:37:10	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned
✓ 44	13:37:41	SHOW CREATE VIEW EmployeeDepartmentView	1 row(s) returned
✓ 45	13:38:30	SHOW FULL TABLES WHERE Table_type = 'VI...	10 row(s) returned

View 14 – Drop a View

Question: Remove an existing View from the database.

DROP VIEW IF EXISTS SalaryRangeView;

The screenshot shows a SQL IDE interface. The top window, titled 'Query 1', contains the SQL command: `DROP VIEW IF EXISTS SalaryRangeView;`. The bottom window, titled 'Output', displays a table of execution results. The table has four columns: '#', 'Time', 'Action', and 'Message'. It contains six rows of data, each with a green checkmark in the first column. The last row shows the execution of the `DROP VIEW IF EXISTS SalaryRangeView;` command, which resulted in '0 row(s) affected'.

#	Time	Action	Message
✓ 41	13:36:40	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
✓ 42	13:37:10	CREATE OR REPLACE VIEW EmployeeBasicVie...	0 row(s) affected
✓ 43	13:37:10	SELECT * FROM EmployeeBasicView LIMIT 0, 1...	5 row(s) returned
✓ 44	13:37:41	SHOW CREATE VIEW EmployeeDepartmentView	1 row(s) returned
✓ 45	13:38:30	SHOW FULL TABLES WHERE Table_type = 'VI...	10 row(s) returned
✓ 46	13:39:07	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected

4. Important Concept: Simple View vs Complex View

A simple View is generally based on one table and does not use grouping or aggregate functions. Some simple Views can be updated. A View containing JOINS, GROUP BY, aggregate functions such as AVG(), DISTINCT, or other non-updatable constructs may not be directly updatable.

5. View vs Table

Feature	Table	View
Stores data	Yes	Normally no; stores query definition
Created with CREATE TABLE	Yes	No
Created with CREATE VIEW	No	Yes
Can use JOINS	Not applicable	Yes
Reflects current underlying data	Own stored data	Yes
Can always be updated	Yes	No; depends on the View

6. Student Practice Tasks

1. Create a View showing only emp_id, emp_name and salary.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS PracticeEmployeeView;
2
3 • CREATE VIEW PracticeEmployeeView AS
4   SELECT emp_id, emp_name, salary
5   FROM employees;
6
7 • SELECT * FROM PracticeEmployeeView;

```

Automatically disabled. manually get caret position and auto

Result Grid

emp_id	emp_name	salary
101	Rahul	60000.00
102	Priya	67000.00
103	Anil	57000.00
104	Sneha	70000.00
105	Kiran	48000.00

Result Grid

Form Editor

PracticeEmployeeView 19 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
✓ 45	13:38:30	SHOW FULL TABLES WHERE Table_type = 'VI...	10 row(s) returned
✓ 46	13:39:07	DROP VIEW IF EXISTS SalaryRangeView	0 row(s) affected
✓ 47	13:39:33	SHOW FULL TABLES WHERE Table_type = 'VI...	9 row(s) returned
⚠ 48	13:41:39	DROP VIEW IF EXISTS PracticeEmployeeView	0 row(s) affected, 1 warning(s): 1051 Un
✓ 49	13:41:39	CREATE VIEW PracticeEmployeeView AS SELE...	0 row(s) affected
✓ 50	13:41:39	SELECT * FROM PracticeEmployeeView LIMIT 0,...	5 row(s) returned

2. Create a View for employees earning more than 60000.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS PracticeHighSalaryView;
2
3 CREATE VIEW PracticeHighSalaryView AS
4 SELECT emp_id, emp_name, salary
5 FROM employees
6 WHERE salary > 60000;
7
8 • SELECT * FROM PracticeHighSalaryView;

```

Automatically disabled. manually get caret position auto

Result Grid

emp_id	emp_name	salary
102	Priya	67000.00
104	Sneha	70000.00

Form Editor

PracticeHighSalaryView 20 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
48	13:41:39	DROP VIEW IF EXISTS PracticeEmployeeView	0 row(s) affected, 1 warning(s): 1051 Un
49	13:41:39	CREATE VIEW PracticeEmployeeView AS SELE...	0 row(s) affected
50	13:41:39	SELECT * FROM PracticeEmployeeView LIMIT 0,...	5 row(s) returned
51	13:42:09	DROP VIEW IF EXISTS PracticeHighSalaryView	0 row(s) affected, 1 warning(s): 1051 Un
52	13:42:09	CREATE VIEW PracticeHighSalaryView AS SEL...	0 row(s) affected
53	13:42:09	SELECT * FROM PracticeHighSalaryView LIMIT ...	2 row(s) returned

3. Create a View showing employee name, department name and salary using JOIN.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS PracticeEmployeeDepartmentView;
2
3 CREATE VIEW PracticeEmployeeDepartmentView AS
4 SELECT
5     e.emp_name,
6     d.dept_name,
7     e.salary
8 FROM employees e
9 JOIN departments d
10     ON e.dept_id = d.dept_id;
11
12 SELECT * FROM PracticeEmployeeDepartmentView;

```

Result Grid

	emp_name	dept_name	salary
▶	Rahul	CSE	60000.00
	Priya	EEE	67000.00
	Anil	CSE	57000.00
	Sneha	EEE	70000.00
	Kiran	ECE	48000.00

PracticeEmployeeDepartmentVi... x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
51	13:42:09	DROP VIEW IF EXISTS PracticeHighSalaryView	0 row(s) affected, 1 warning(s): 1051 Unl
52	13:42:09	CREATE VIEW PracticeHighSalaryView AS SEL...	0 row(s) affected
53	13:42:09	SELECT * FROM PracticeHighSalaryView LIMIT ...	2 row(s) returned
54	13:42:39	DROP VIEW IF EXISTS PracticeEmployeeDepart...	0 row(s) affected, 1 warning(s): 1051 Unl
55	13:42:39	CREATE VIEW PracticeEmployeeDepartment Vie...	0 row(s) affected
56	13:42:39	SELECT * FROM PracticeEmployeeDepartment Vi...	5 row(s) returned

4. Create a View showing only ECE employees.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS PracticeECEEmployees;
2
3 CREATE VIEW PracticeECEEmployees AS
4 SELECT
5     e.emp_id,
6     e.emp_name,
7     d.dept_name,
8     e.salary
9 FROM employees e
10 JOIN departments d
11     ON e.dept_id = d.dept_id
12 WHERE d.dept_name = 'ECE';

```

Result Grid

emp_id	emp_name	dept_name	salary
105	Kiran	ECE	48000.00

Form Editor

PracticeECEEmployees 22 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
54	13:42:39	DROP VIEW IF EXISTS PracticeEmployeeDepart...	0 row(s) affected, 1 warning(s): 1051 Unl
55	13:42:39	CREATE VIEW PracticeEmployeeDepartment Vie...	0 row(s) affected
56	13:42:39	SELECT * FROM PracticeEmployeeDepartment Vi...	5 row(s) returned
57	13:43:05	DROP VIEW IF EXISTS PracticeECEEmployees	0 row(s) affected, 1 warning(s): 1051 Unl
58	13:43:05	CREATE VIEW PracticeECEEmployees AS SELE...	0 row(s) affected
59	13:43:05	SELECT * FROM PracticeECEEmployees LIMIT 0...	1 row(s) returned

5. Create a View showing department-wise employee count.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS PracticeDepartmentCount;
2
3 CREATE VIEW PracticeDepartmentCount AS
4 SELECT
5     d.dept_name,
6     COUNT(e.emp_id) AS EmployeeCount
7 FROM employees e
8 JOIN departments d
9     ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name;
11
12 • SELECT * FROM PracticeDepartmentCount;

```

Result Grid | Filter Rows: | Export: | Write

	dept_name	EmployeeCount
▶	CSE	2
	EEE	2
	ECE	1

Result Grid
Form Editor

PracticeDepartmentCount 23 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
57	13:43:05	DROP VIEW IF EXISTS PracticeECEEmployees	0 row(s) affected, 1 warning(s): 1051 Un
58	13:43:05	CREATE VIEW PracticeECEEmployees AS SELE...	0 row(s) affected
59	13:43:05	SELECT * FROM PracticeECEEmployees LIMIT 0...	1 row(s) returned
60	13:43:31	DROP VIEW IF EXISTS PracticeDepartmentCount	0 row(s) affected, 1 warning(s): 1051 Un
61	13:43:31	CREATE VIEW PracticeDepartmentCount AS SE...	0 row(s) affected
62	13:43:31	SELECT * FROM PracticeDepartmentCount LIM...	3 row(s) returned

6. Create a View showing department-wise maximum salary.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS PracticeDepartmentMaxSalary;
2
3 CREATE VIEW PracticeDepartmentMaxSalary AS
4 SELECT
5     d.dept_name,
6     MAX(e.salary) AS MaximumSalary
7 FROM employees e
8 JOIN departments d
9     ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name;
11
12 SELECT * FROM PracticeDepartmentMaxSalary;

```

Result Grid

	dept_name	MaximumSalary
▶	CSE	60000.00
	EEE	70000.00
	ECE	48000.00

PracticeDepartmentMaxSalary 24 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
60	13:43:31	DROP VIEW IF EXISTS PracticeDepartmentCount	0 row(s) affected, 1 warning(s): 1051 Un
61	13:43:31	CREATE VIEW PracticeDepartmentCount AS SE...	0 row(s) affected
62	13:43:31	SELECT * FROM PracticeDepartmentCount LIM...	3 row(s) returned
63	13:44:01	DROP VIEW IF EXISTS PracticeDepartmentMax...	0 row(s) affected, 1 warning(s): 1051 Un
64	13:44:01	CREATE VIEW PracticeDepartmentMaxSalary A...	0 row(s) affected
65	13:44:01	SELECT * FROM PracticeDepartmentMaxSalary ...	3 row(s) returned

7. Create a View showing departments whose average salary is greater than 50000.

Query 1 x

SQLAdditions

Limit to 1000 rows

```

1 DROP VIEW IF EXISTS PracticeHighAverageDepartments;
2
3 CREATE VIEW PracticeHighAverageDepartments AS
4 SELECT
5     d.dept_name,
6     AVG(e.salary) AS AverageSalary
7 FROM employees e
8 JOIN departments d
9     ON e.dept_id = d.dept_id
10 GROUP BY d.dept_id, d.dept_name
11 HAVING AVG(e.salary) > 50000;
12

```

Automatically disabled. manually get caret position.

Result Grid

	dept_name	AverageSalary
▶	CSE	58500.000000
	EEE	68500.000000

Export: Wra

Result Grid

Form Editor

PracticeHighAverageDepartmen... x

Read Only

Context Help Snip

Output

Action Output

#	Time	Action	Message
63	13:44:01	DROP VIEW IF EXISTS PracticeDepartmentMax...	0 row(s) affected, 1 warning(s): 1051 Unl
64	13:44:01	CREATE VIEW PracticeDepartmentMaxSalary A...	0 row(s) affected
65	13:44:01	SELECT * FROM PracticeDepartmentMaxSalary ...	3 row(s) returned
66	13:44:26	DROP VIEW IF EXISTS PracticeHighAverageDe...	0 row(s) affected, 1 warning(s): 1051 Unl
67	13:44:26	CREATE VIEW PracticeHighAverageDepartment...	0 row(s) affected
68	13:44:26	SELECT * FROM PracticeHighAverageDepartme...	2 row(s) returned

8. Create a View with employee name, salary and annual salary.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1 • DROP VIEW IF EXISTS PracticeAnnualSalary;
2
3 CREATE VIEW PracticeAnnualSalary AS
4 SELECT
5     emp_name,
6     salary,
7     salary * 12 AS AnnualSalary
8 FROM employees;
9
10 • SELECT * FROM PracticeAnnualSalary;

```

Result Grid | Filter Rows: | Export: | Wrap

	emp_name	salary	AnnualSalary
▶	Rahul	60000.00	720000.00
	Priya	67000.00	804000.00
	Anil	57000.00	684000.00
	Sneha	70000.00	840000.00
	Kiran	48000.00	576000.00

Result Grid
Form Editor

PracticeAnnualSalary 26 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
66	13:44:26	DROP VIEW IF EXISTS PracticeHighAverageDe...	0 row(s) affected, 1 warning(s): 1051 Unl
67	13:44:26	CREATE VIEW PracticeHighAverageDepartment...	0 row(s) affected
68	13:44:26	SELECT * FROM PracticeHighAverageDepartme...	2 row(s) returned
69	13:44:56	DROP VIEW IF EXISTS PracticeAnnualSalary	0 row(s) affected, 1 warning(s): 1051 Unl
70	13:44:56	CREATE VIEW PracticeAnnualSalary AS SELEC...	0 row(s) affected
71	13:44:56	SELECT * FROM PracticeAnnualSalary LIMIT 0, ...	5 row(s) returned

9. Update an employee's salary through a simple View and verify the base table.

Query 1 x SQLAdditions

Limit to 1000 rows

```

4 SELECT emp_id, emp_name, salary
5 FROM employees;
6
7 • UPDATE PracticeSalaryUpdateView
8 SET salary = 62000
9 WHERE emp_id = 101;
10
11 • SELECT * FROM PracticeSalaryUpdateView
12 WHERE emp_id = 101;
13
14 • SELECT * FROM employees
15 WHERE emp_id = 101;

```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	62000.00
*	NULL	NULL	NULL	NULL

Result Grid

Form Editor

PracticeSalaryUpdateView 27 employees 28 x Apply Revert Context Help Snip

Output

Action Output

#	Time	Action	Message
✓ 71	13:44:56	SELECT * FROM PracticeAnnualSalary LIMIT 0, ...	5 row(s) returned
⚠ 72	13:45:24	DROP VIEW IF EXISTS PracticeSalaryUpdateView	0 row(s) affected, 1 warning(s): 1051 Unl
✓ 73	13:45:24	CREATE VIEW PracticeSalaryUpdateView AS S...	0 row(s) affected
✓ 74	13:45:24	UPDATE PracticeSalaryUpdateView SET salary =...	1 row(s) affected Rows matched: 1 Cha
✓ 75	13:45:24	SELECT * FROM PracticeSalaryUpdateView WH...	1 row(s) returned
✓ 76	13:45:24	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned

10. Use SHOW CREATE VIEW to display the definition of a View.

Query 1 x SQLAdditions

Limit to 1000 rows

1 • SHOW CREATE VIEW PracticeSalaryUpdateView;

Result Grid Filter Rows: Export: Wrap Cell

View	Create View
practicesalaryupdateview	CREATE ALGORITHM=UNDEFINED DEFINER=

Result Grid Form Editor

Result 29 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
72	13:45:24	DROP VIEW IF EXISTS PracticeSalaryUpdateView	0 row(s) affected, 1 warning(s): 1051 Un
73	13:45:24	CREATE VIEW PracticeSalaryUpdateView AS S...	0 row(s) affected
74	13:45:24	UPDATE PracticeSalaryUpdateView SET salary =...	1 row(s) affected Rows matched: 1 Cha
75	13:45:24	SELECT * FROM PracticeSalaryUpdateView WH...	1 row(s) returned
76	13:45:24	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
77	13:45:53	SHOW CREATE VIEW PracticeSalaryUpdateView	1 row(s) returned

11. List all Views in the database.

Query 1 x SQLAdditions

Limit to 1000 rows

```

1  SHOW FULL TABLES
2  WHERE Table_type = 'VIEW';

```

Result Grid | Filter Rows: | Export: | Wrap Cells

Tables_in_employeeodb1	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
hinhsalaryemployees	VTFW

Result 30 x Read Only Context Help Snip

Output

Action Output

#	Time	Action	Message
73	13:45:24	CREATE VIEW PracticeSalaryUpdateView AS S...	0 row(s) affected
74	13:45:24	UPDATE PracticeSalaryUpdateView SET salary =...	1 row(s) affected Rows matched: 1 Cha
75	13:45:24	SELECT * FROM PracticeSalaryUpdateView WH...	1 row(s) returned
76	13:45:24	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
77	13:45:53	SHOW CREATE VIEW PracticeSalaryUpdateView	1 row(s) returned
78	13:46:45	SHOW FULL TABLES WHERE Table_type = 'VI...	18 row(s) returned

12. Drop a View and verify that it no longer appears in the list.

Query 1



Limit to 1000 rows

1

`DROP VIEW IF EXISTS PracticeSalaryUpdateView;`

SQLAdditions

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Context Help

Snip

Output

Action Output

#	Time	Action	Message
✓ 74	13:45:24	UPDATE PracticeSalaryUpdateView SET salary =...	1 row(s) affected Rows matched: 1 Cha
✓ 75	13:45:24	SELECT * FROM PracticeSalaryUpdateView WH...	1 row(s) returned
✓ 76	13:45:24	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
✓ 77	13:45:53	SHOW CREATE VIEW PracticeSalaryUpdateView	1 row(s) returned
✓ 78	13:46:45	SHOW FULL TABLES WHERE Table_type = 'VI...	18 row(s) returned
✓ 79	13:47:13	DROP VIEW IF EXISTS PracticeSalaryUpdateView	0 row(s) affected

Query 1 x

SQLAdditions

Limit to 1000 rows

```

1 SHOW FULL TABLES
2 WHERE Table_type = 'VIEW';

```

Result Grid

Tables_in_employeedb1	Table_type
annualsalaryview	VIEW
cseemployees	VIEW
departmentsalaryview	VIEW
employeebasicview	VIEW
employeedepartmentview	VIEW
employeesalaryview	VIEW
employeeview	VIEW
highaveragedepartments	VIEW
highsalaryemployees	VIEW

Result 31 x

Read Only

Context Help

Output

Action Output

#	Time	Action	Message
75	13:45:24	SELECT * FROM PracticeSalaryUpdateView WH...	1 row(s) returned
76	13:45:24	SELECT * FROM employees WHERE emp_id = 1...	1 row(s) returned
77	13:45:53	SHOW CREATE VIEW PracticeSalaryUpdateView	1 row(s) returned
78	13:46:45	SHOW FULL TABLES WHERE Table_type = 'VI...	18 row(s) returned
79	13:47:13	DROP VIEW IF EXISTS PracticeSalaryUpdateView	0 row(s) affected
80	13:47:42	SHOW FULL TABLES WHERE Table_type = 'VI...	17 row(s) returned

7. Viva / Discussion Questions

1. What is a View?

A View is a **virtual table** created from one or more existing tables. It stores the **SQL query/definition**, not a separate copy of the data.

2. Why do we use Views?

Views are used to **simplify complex queries**, provide easier access to selected data, and help with **security** by exposing only required columns or rows.

3. Does a View normally store a separate copy of the data?

No. A View normally stores only the **query definition**. When we query the View, MySQL retrieves the current data from the underlying tables.

4. What is the difference between a View and a table?

A **Table** normally stores the actual data, while a **View** normally stores a query definition and displays data from underlying tables.

5. Can a View be created using multiple tables?

Answer:

Yes. A View can be created using multiple tables, usually with JOIN.

For example, our EmployeeDepartmentView used:

```
FROM employees e  
JOIN departments d  
ON e.dept_id = d.dept_id;
```

6. Can every View be updated?

No. It depends on the View.

Simple Views based on a single table may be updatable, while Views containing things like **JOIN**, **GROUP BY**, **aggregate functions**, or **other non-updatable constructs** may not be directly updatable.

7. Why are Views useful for security?

Views can expose **only the required rows or columns** instead of giving users access to the entire underlying table.

8. What is the difference between CREATE VIEW and CREATE OR REPLACE VIEW?

Answer:

CREATE VIEW creates a **new View**.

CREATE OR REPLACE VIEW modifies an **existing View** or creates it if appropriate.

9. How do you display the definition of a View in MySQL?

Use:

```
SHOW CREATE VIEW view_name;
```

10. How do you remove a View?

Use:

```
DROP VIEW IF EXISTS view_name;
```

8. Common Errors and Solutions

Table does not exist: Run USE employeeadb1; and check with SHOW TABLES;.

View already exists: Use DROP VIEW IF EXISTS view_name; or CREATE OR REPLACE VIEW.

Unknown column: Check the column names using DESCRIBE employees; or DESCRIBE departments;.

Cannot update View: The View may contain JOIN, GROUP BY, aggregate functions or another non-updatable construct.

Wrong database: Check the selected schema and run SELECT DATABASE();.

9. Quick Reference

```
CREATE VIEW view_name AS  
SELECT ...;
```

```
SELECT * FROM view_name;
```

```
CREATE OR REPLACE VIEW view_name AS  
SELECT ...;
```

```
SHOW CREATE VIEW view_name;
```

```
SHOW FULL TABLES  
WHERE Table_type='VIEW';
```

```
DROP VIEW IF EXISTS view_name;
```